Progress and Preservation

Concepts of Programming Languages

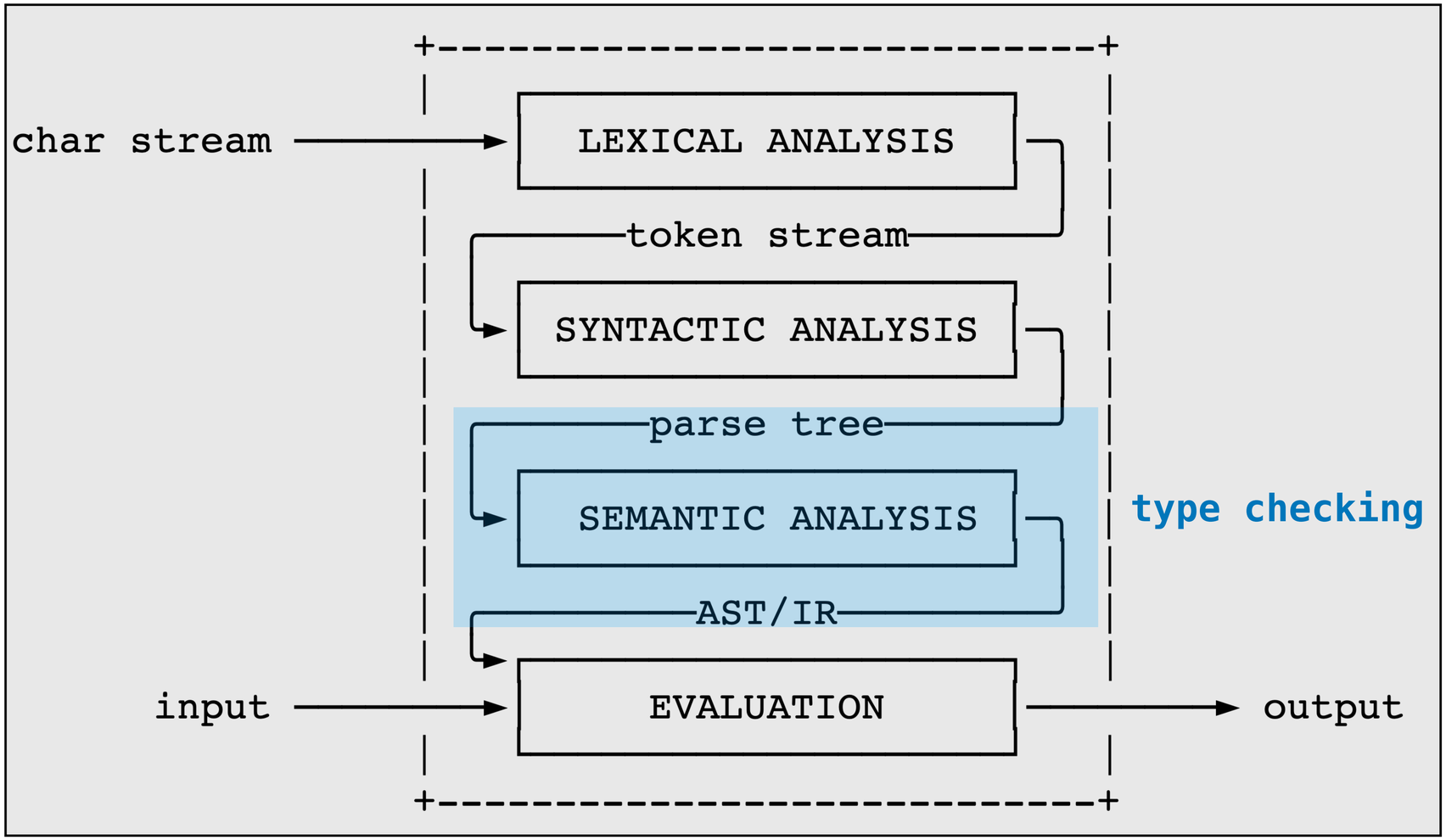
CAS CS 320

Outline

- Discuss **induction** over derivations
- Show that STLC satisfies **progress** and **preservation**

Recap

Recall: The Picture



Recall: Type Checking/Inference

```
type_check : expr -> ty -> bool
type_of : expr -> ty option
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Type inference: *synthesizing* a type for an expression

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type_check : expr -> ty -> bool

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Type checking: determining whether an expression can be typed

Type inference: *synthesizing* a type for an expression

Theoretically, these two problems can be very different. *For STLC, they are both easy*

Recall: Syntax (STLC)

type annotations

satisfies uniqueness of type

$$e ::= \bullet \mid x \mid \lambda x^{\boxed{\tau}}.\, e \mid ee$$

$$\tau ::= \top \mid \tau \to \tau$$

$$x ::= variables$$

The syntax is the same as that of the lambda calculus except:

- » we include a unit expression
- » we have types, which annotate arguments

Recall: Typing (STLC)

$$\frac{}{\Gamma \vdash \bullet : \top}\ \texttt{unit}$$

$$\frac{\Gamma, x : \tau \vdash e : \tau'}{\Gamma \vdash \lambda x^{\tau}.\, e : \tau \to \tau'}\ \texttt{abstraction}$$

$$\frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau}\ \texttt{variable}$$

$$\frac{\Gamma \vdash e_1 : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \tau'}\ \texttt{application}$$

These rules enforce that a function can only be applied if we *know* that it's a function

Recall: Semantics (STLC)

$$\frac{e_1 \longrightarrow e_1'}{e_1 e_2 \longrightarrow e_1' e_2}\ \texttt{leftEval}$$

$$\frac{}{(\lambda x\,.\,e)e' \longrightarrow [e'/x]e}\ \texttt{beta}$$

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We can use any semantics (this is small-step CBN)

This is part of the point. Type-checking only determines *whether* we go on to evaluate the program

It doesn't determine *how* we evaluate the program

Practice Problem

$\Gamma \vdash e : \tau$

find this

and this

$FV(e) = \{w, y\}$

$$\overset{e}{(\lambda x^{(\mathsf{T}\to\mathsf{T})\to\mathsf{T}}.\, x(\lambda z^{\mathsf{T}}.\, x(w\, z)))\, y}$$

$\{y : \tau_1, w : \tau_2\} \vdash e : \tau$

Determine the smallest context such that the above expression is typeable (also give its type)

Practice Problem

$$(\lambda x^{(\top \to \top) \to \top} . x(\lambda z^{\top} . x(wz)))y$$

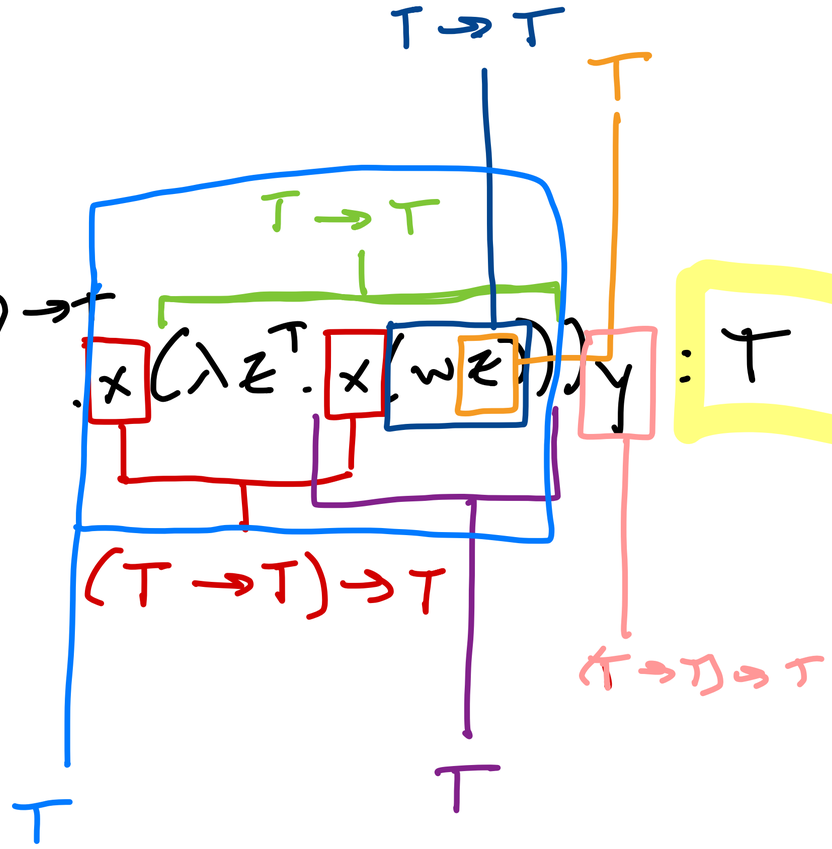
Determine the smallest context such that the above expression is typeable (also give its type)

Answer

$(\lambda \overset{\boxed{A}}{x} e_1)\boxed{e_2}$

$$(\lambda x^{(\mathsf{T}\to\mathsf{T})\to\mathsf{T}}.x(\lambda z^{\mathsf{T}}.x(wz)))y$$

$$\{y : (T\to T)\to T,\ w : T\to(T\to T)\} \vdash (\lambda x^{(T\to T)\to T}.x(\lambda z^{T}.x(wz)))y : T$$



Induction over Derivations

The Key Idea

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$$\dfrac{\dfrac{}{\{\} \vdash \texttt{2} : \texttt{int}}\,(\text{intLit}) \qquad \dfrac{\dfrac{}{\{\texttt{y} : \texttt{int}\} \vdash \texttt{y} : \texttt{int}}\,(\text{var}) \qquad \dfrac{}{\{\texttt{y} : \texttt{int}\} \vdash \texttt{y} : \texttt{int}}\,(\text{var})}{\{\texttt{y} : \texttt{int}\} \vdash \texttt{y} + \texttt{y} : \texttt{int}}\,(\text{intAdd})}{\{\} \vdash \texttt{let y = 2 in y} + \texttt{y} : \texttt{int}}\,(\text{let})$$

Derivations are *trees*

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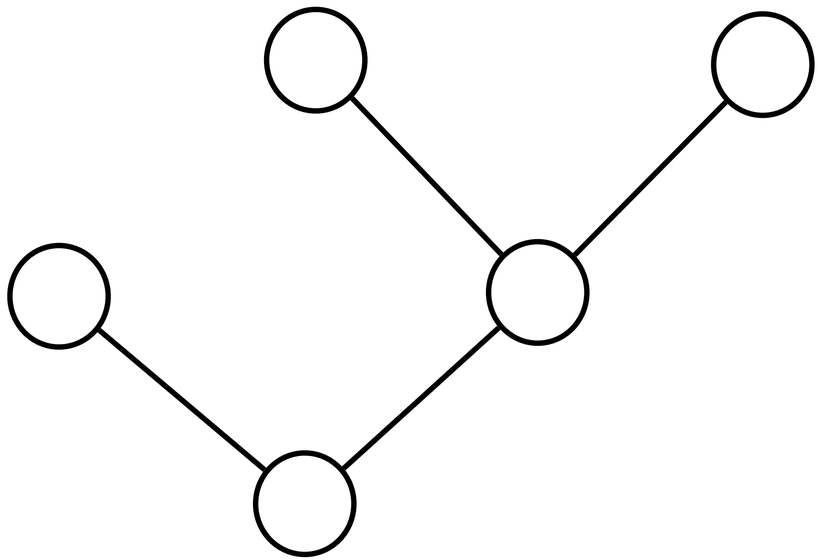
Derivations are *trees*

We can prove things about trees using induction



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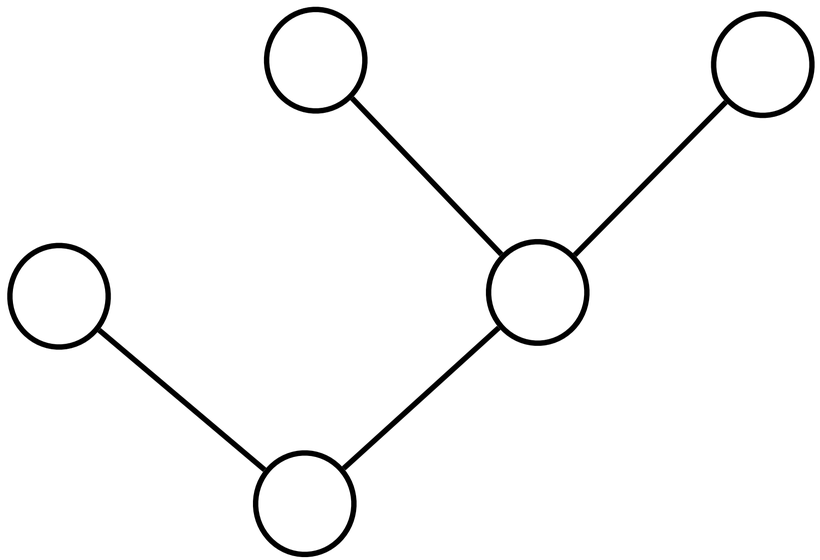
We can prove things about *derivable judgments* using induction



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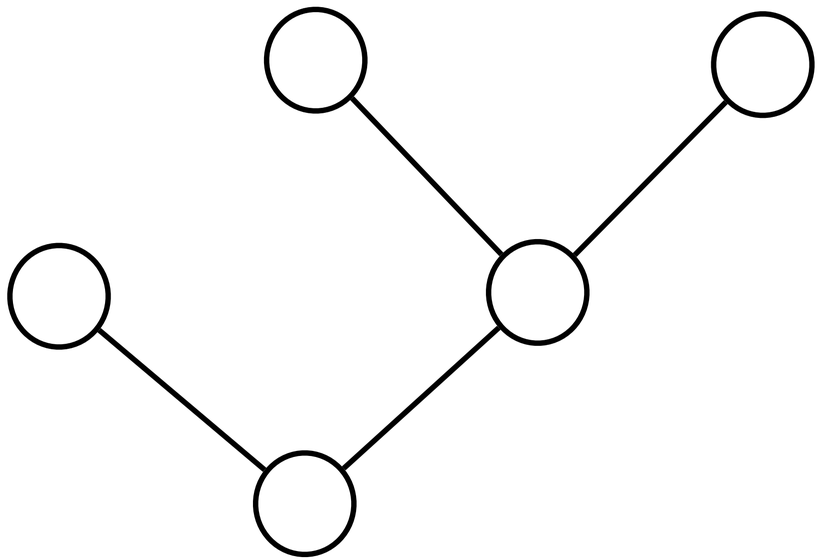
Derivations are *trees*

We can prove things about trees using induction

We can prove things about *derivable judgments* using induction

Important: Every derivable judgment corresponds to a derivation



Warm-up: Binary Trees

```
type 'a tree =
  | Empty
  | Node of 'a tree * 'a * 'a tree
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Let $\mathsf{size}(T)$ denote the number of Nodes and let $\mathsf{height}(T)$ denote the length of the *longest* path from the root to Empty

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Theorem. $\text{size}(T) \leq 2^{\text{height}(T)} - 1$ for any tree T

Proof. By induction on the *structure of trees*

Base Case: Empty

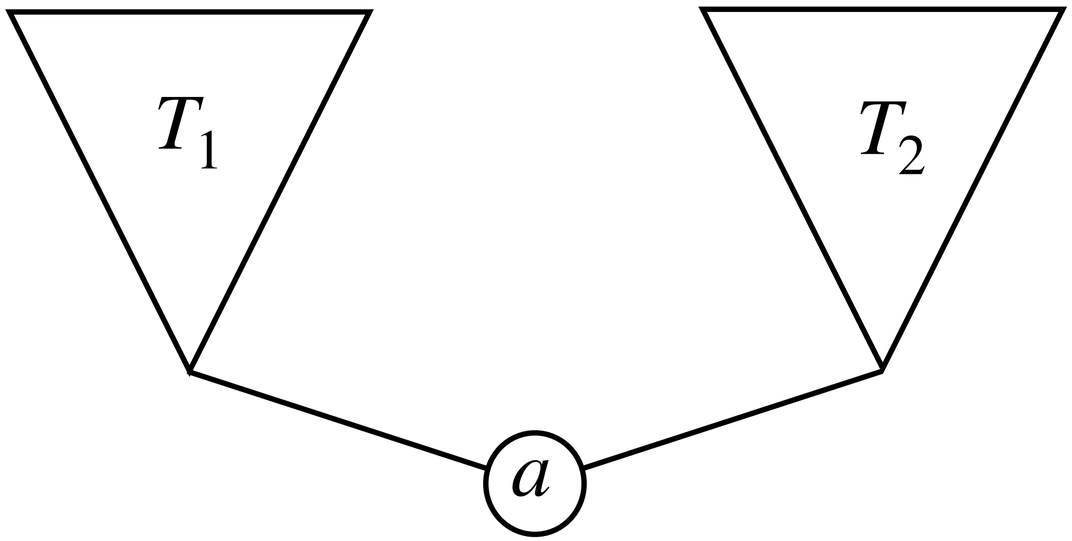
$$\text{size}(\emptyset) \leq 2^{h(\emptyset)} - 1$$

$$\text{size}(\emptyset) = 0 \qquad h(\emptyset) = 0$$

$$0 \leq 2^0 - 1 = 0$$

✓

Inductive Hypothesis



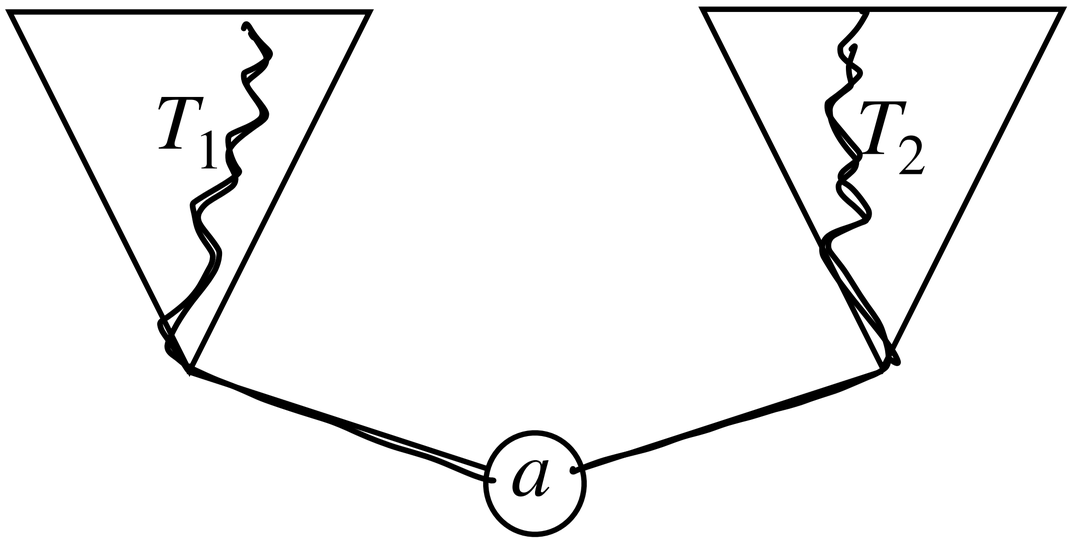
If T is of the form `Node` $(T_1,\ a,\ T_2)$ then $\text{size}(T_1) \leq 2^{\text{height}(T_1)} - 1$ and $\text{size}(T_2) \leq 2^{\text{height}(T_2)} - 1$

That is, we get to assume that what we want holds of our subtrees

Inductive Step: Nodes

WTS:

$$2^{h(T_1)} + 2^{h(T_2)} \le 2^{h(T)}$$



$$\begin{aligned}
\text{size}(T) &= 1 + \text{size}(T_1) + \text{size}(T_2) \quad \text{(by def.)} \\
&\le 1 + 2^{\text{height}(T_1)} - 1 + 2^{\text{height}(T_2)} - 1 \quad \text{(by IH)} \\
&= 2^{h(T_1)} + 2^{h(T_2)} - 1 \le \left(2^{h(T_1)-1}\right) \cdot 2 - 1 = 2^{h(T)} - 1
\end{aligned}$$

Another Warm-up: Well-Scopedness

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An expression e is **well-scoped** with respect to a context Γ if $x \in FV(e)$ implies x appears in Γ

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Another Warm-up: Well-Scopedness

An expression e is **well-scoped** with respect to a context Γ if $x \in FV(e)$ implies x appears in Γ

$\Gamma \vdash e : \tau$

Theorem. If e is well-typed in Γ, then e is well-scoped

Proof. By induction on *derivations*

Base Case: Axioms

$$\frac{}{\Gamma \vdash \bullet : \top}\ \textbf{unit}$$

$FV(\bullet) = \emptyset$ ✓

$$\frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau}\ \textbf{variable}$$

$FV(x) \overset{?}{\in} \Gamma$ ✓

yes, bec. sid. cond. says so.

We need to show that expressions typed using just axioms satisfy well-scopedness

Inductive Hypothesis

$$\frac{\begin{array}{c}\vdots\\ \mathscr{D}_1\\ \Gamma_1 \vdash e_1 : \tau_1\end{array} \quad \begin{array}{c}\vdots\\ \mathscr{D}_2\\ \Gamma_2 \vdash e_2 : \tau_2\end{array} \quad \ldots \quad \begin{array}{c}\vdots\\ \mathscr{D}_k\\ \Gamma_k \vdash e_k : \tau_k\end{array}}{\Gamma \vdash e : \tau}$$

If $e_1, \ldots, e_k$ *are well-scoped (because they are typeable in the each of their contexts)*

Inductive Step 1: Application

IH:

$FV(e_1) \subseteq \Gamma$

$$\frac{\begin{array}{c}\vdots \\ \mathscr{D}_1 \\ \Gamma \vdash e_1 : \tau \to \tau'\end{array} \qquad \begin{array}{c}\vdots \\ \mathscr{D}_2 \\ \Gamma \vdash e_2 : \tau\end{array}}{\Gamma \vdash e_1 e_2 : \tau'}\ \textbf{application}$$

$FV(e_2) \subseteq \Gamma$

$FV(e_1 e_2) \overset{?}{\subseteq} \Gamma$

$FV(e_1 e_2) = FV(e_1) \cup FV(e_2) \subseteq \Gamma$ (by IH)

What if the last rule I applied was application?

Inductive Step 2: Abstraction

$\lambda x . x$

$$\frac{\begin{array}{c}\vdots \\ \mathscr{D} \\ \Gamma, x : \tau \vdash e : \tau'\end{array}}{\Gamma \vdash \lambda x^{\tau} . e : \tau \to \tau'}\ \textbf{abstraction}$$

$FV(\lambda x^{\tau}.e) = FV(e) \setminus \{x\} \subseteq \Gamma$

by IH, $FV(e) \subseteq \Gamma, x : \tau \Rightarrow FV(e) \setminus \{x\} \subseteq \Gamma$ ✓

What if the last rule I applied was abstraction?

Progress and Preservation

Recall: Type Safety

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Thm. If $\cdot \vdash e : \tau$ and there is a value v such that $\langle \varnothing, e \rangle \Downarrow v$ then $\cdot \vdash e(v) : \tau$

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With small-step semantics, we can give a finer-grained analysis:

Recall: Type Safety

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With small-step semantics, we can give a finer-grained analysis:

Thm. If $\cdot \vdash e : \tau$, then

doesn't imply termination

» *(progress)* either e is a value or there is an e' such that $e \longrightarrow e'$

» *(preservation)* If $\cdot \vdash e : \tau$ and $e \longrightarrow e'$ then $\cdot \vdash e' : \tau$

no stuck terms

$$(\lambda x.\ \lambda y.\ x + y)\ 2\ 3 \longrightarrow (\lambda y.\ 2 + y)\ 3$$

$$\longrightarrow 2 + 3 \longrightarrow 5$$

Recall: Type Safety

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With small-step semantics, we can give a finer-grained analysis:

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> » *(progress)* either e is a value or there is an e' such that $e \longrightarrow e'$
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These results are *fundamental*. They tell us that our PL is well-behaved (it's a "good" PL)

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With small-step semantics, we can give a finer-grained analysis:

goal for today

> **Thm.** If $\cdot \vdash e : \tau$, then
>
> » *(progress)* either e is a value or there is an e' such that $e \longrightarrow e'$
>
> » *(preservation)* If $\cdot \vdash e : \tau$ and $e \longrightarrow e'$ then $\cdot \vdash e' : \tau$

These results are *fundamental*. They tell us that our PL is well-behaved (it's a "good" PL)

Disclaimer: We're gonna hand-wave liberally

Recall: STLC

Syntax

$$e ::= \bullet \mid x \mid \lambda x^{\tau}.\, e \mid ee$$

$$\tau ::= \top \mid \tau \to \tau$$

$$x ::= \mathit{variables}$$

Typing

$$\frac{}{\Gamma \vdash \bullet : \top}\ \texttt{unit}$$

$$\frac{\Gamma, x : \tau \vdash e : \tau'}{\Gamma \vdash \lambda x^{\tau}.\, e : \tau \to \tau'}\ \texttt{abstraction}$$

$$\frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau}\ \texttt{variable}$$

$$\frac{\Gamma \vdash e_1 : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \tau'}\ \texttt{application}$$

Semantics

$$\frac{e_1 \longrightarrow e_1'}{e_1 e_2 \longrightarrow e_1' e_2}\ \texttt{leftEval}$$

$$\frac{}{(\lambda x.\, e)e' \longrightarrow [e'/x]e}\ \texttt{beta}$$

Progress (STLC)

$\emptyset \vdash e : \tau$

Theorem. If e is well-typed then e is a value or there is an expression e' such that $e \longrightarrow e'$

Proof. By induction over *derivations*

Base Case: Axioms

$$\frac{}{\cdot \vdash \bullet : \top}\ \texttt{unit}$$

already a value

$$\frac{(x : \tau) \in \varnothing}{\underbrace{\cdot \vdash x : \tau}}\ \texttt{variable}$$

not possible by well-scoped

We need to show that expressions typed using just axioms yield non-stuck terms

Inductive Step 1: Application

$$\frac{e_1 \to e_1'}{e_1 e_2 \to e_1' e_2}\ \text{(lem)}$$

$$\frac{\cdot \vdash e_1 : \tau \to \tau' \qquad \cdot \vdash e_2 : \tau}{\cdot \vdash e_1 e_2 : \tau'}\ \textbf{application}$$

if e_1 is a value:

(i) $\bullet$ or (ii) $\lambda x.e$

(i) if $\Gamma \vdash \bullet : \tau \Rightarrow \tau = T$ by type uniqueness, so not possible because

(ii)

$$\frac{}{(\lambda x.e)\, e_2 \to [e_2/x]e}\ \text{(beta)}$$

$$\frac{\vdots}{e_1 \to e_1'}$$

if e_1 is not a value: by IH since $\cdot \vdash e_1 : \tau \to \tau \Rightarrow \exists e_1'$ s.t.

*What do we know given that e_1 is either a **value** or **reducible**?*

Inductive Step 2: Abstraction

value:

- or $\lambda x^{\tau}.e$

$$\frac{\{x : \tau\} \vdash e : \tau'}{\cdot \vdash \underbrace{\lambda x^{\tau}.e}_{\text{already a value.}} : \tau \to \tau'} \text{ abstraction}$$

Our expression already a value if the last rule we applied was abstraction!

Preservation (STLC)

Theorem. If $\Gamma \vdash e : \tau$ and $e \longrightarrow e'$ then $\Gamma \vdash e' : \tau$

Proof. By induction over *derivations*

This one is much tricker...

Base Case: Axioms

$$\frac{}{\Gamma \vdash \bullet : \top}\ \texttt{unit}$$

$$\frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau}\ \texttt{variable}$$

Expressions typed using just axioms cannot be reduced (nothing to do here)

Inductive Step 1: Abstraction

$$\frac{\Gamma, x : \tau \vdash e : \tau'}{\Gamma \vdash \lambda x^{\tau} . e : \tau \to \tau'} \text{ abstraction}$$

Expressions derived using abstraction as the last rule is already a value (nothing to do here)

Inductive Step 2: Application

$$\frac{\Gamma \vdash e_1 : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \tau'} \text{ application}$$

This is where the work comes in...

The trick: We do induction (inside our current induction) on the structure of *semantic* derivations!

What possible ways can $e_1 e_2$ *be reduced?*

Inductive Step 2.1: leftEval

$$\frac{e_1 \longrightarrow e_1'}{e_1 e_2 \longrightarrow e_1' e_2}\ \texttt{leftEval} \qquad \frac{\Gamma \vdash e_1 : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \boxed{\tau'}}\ \texttt{application}$$

$\Gamma \vdash e_1 : \tau \to \tau'$ and $e_1 \to e_1'$ then IH $\Gamma \vdash e_1' : \tau \to \tau'$

so.

$$\frac{\Gamma \vdash e_1' : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1' e_2 : \boxed{\tau'}}\ (\text{app})$$

What if our last rule was an application ***and*** $e_1 e_2$ *is reducible by leftEval?*

Inductive Step 2.1: leftEval

$$\frac{}{(\lambda x\,.\,e)e_2 \longrightarrow [e_2/x]e}\ \textbf{beta} \qquad \frac{\dfrac{\Gamma, x : \tau \vdash e : \tau'}{\Gamma \vdash (\lambda x^{\tau}.\, e) : \tau \to \tau'} \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash (\lambda x^{\tau}.\, e)e_2 : \boxed{\tau'}}\ \textbf{application}$$

$\Gamma, x : \tau \vdash e : \tau'$ and $\Gamma \vdash e_2 : \tau \implies \Gamma \vdash [e_2/x]e : \boxed{\tau'}$

(by subst.)

What if our last rule was an application ***and*** $e_1 e_2$ *is reducible by beta?*

Substitution Lemma

$\cdot \vdash 1 : int$

$\{x : int\} \vdash x = 0 : bool$

$\Downarrow$

$\cdot \vdash 1 = 0 : bool$

Lemma. If $\Gamma \vdash e_2 : \tau_2$ and $\Gamma, x : \tau_2 \vdash e : \tau$ then

$$\Gamma \vdash [e_2/x]e : \tau$$

That is, if e is well-typed in a context with (x, τ) then we can substitute x with anything of type τ and it's still the same type

(we can prove this by, you guessed it, induction on derivations)

$\{ f : int \to bool, \boxed{x : int}, y : int \} \vdash \lambda z^{int}.\ if\ f\ \boxed{x}\ then\ \boxed{x} + y$

$else\ z : int \to int$

$\{ f : int \to bool, y : int \} \vdash \boxed{if\ f\ y\ then\ 0\ else\ 1} : int$

$\vdash \lambda z^{int}.\ if\ f\ \boxed{(if\ f\ y\ then\ 0\ else\ 1)}\ then\ \boxed{(if\ f\ y\ then\ 0\ else\ 1)} + y$

$else\ z : int \to int$

$\{ f : int \to bool, y : int \}$

The Point

```
let rec eval env e =
  match e with
  | Var x -> Env.find x env
  ...
```

Progress and preservation tell us that **terms never get stuck during evaluation**

This is huge. I can't emphasize this enough

Our type system ensures we only evaluate programs that make sense!

Summary

Progress and **preservation** are fundamental features of good programming languages

We can prove things about well-typed expressions by performing **induction** over derivations